



# INDIAN INSTITUTE OF TECHNOLOGY KHARAGPUR

## Mid-Autumn Semester Examination 2022-23

Date of Examination: \_\_\_\_\_ Session: (FN/AN) \_\_\_\_\_ Duration: 2 hrs Full Marks: 40  
Subject No.: MF41601 Subject: Soft Computing  
Department/Center/School: Mechanical Engineering  
Specific charts, graph paper, log book etc., required N.A.  
Special Instructions (if any): Answer all the questions. Assume suitable data, if necessary.

Marks: 10+10+2+4+8+4+2=40

Q.1. Answer all MCQs. There is no negative marking.

(i) Exhaustive search method is applicable to

- (a) Multi-modal objective function
- (b) Unimodal maximization problem only
- (c) Unimodal minimization problem only
- (d) Unimodal either maximization or minimization problem

(ii) The chance of the solutions of steepest descent method for being trapped into the local minima is more as

- (a) Gradient of a function is local property
- (b) it is computationally fast
- (c) its iteration-wise step length can be calculated mathematically
- (d) it is an iterative approach

(iii) According to Building-block Hypothesis of a binary-coded genetic algorithm, a schema will survive through the iterations, if it has

- (a) High order, long defining length and fitness more than average fitness of the population
- (b) High order, long defining length and fitness less than average fitness of the population
- (c) Low order, long defining length and fitness less than average fitness of the population
- (d) Low order, short defining length and fitness more than average fitness of the population

(iv) Let us consider a function of a single variable  $y = f(x) = x^4$ . At the point  $x = x^* = 0$ , it has the

- (a) Maximum point
- (b) Minimum point
- (c) Saddle/inflection point
- (d) Both the maximum and minimum points

(v) Parallel Computing

- (a) can be implemented for traditional (conventional) optimization tools.
- (b) can be implemented for non-traditional (nature-inspired) optimization tools.
- (c) can be implemented for both traditional and non-traditional optimization tools.
- (d) can be implemented for neither traditional nor non-traditional optimization tools.

Q. 2. Let us try to solve a maximization problem of the form:  $y = f(x_1, x_2) = x_1^2 + x_2^2$ , where  $x_1$  and  $x_2$  are two real variables lying in the range of (6.0, 10.0). Use a binary-coded Genetic Algorithm (GA) to solve this maximization problem. Its initial population of size  $N = 4$  created at random is given below. Let us use 4 bits to represent each of the variables, as given below.

1011 1001  
0110 1110  
1010 0101  
1101 0111

Using Roulette-Wheel selection, determine their probability values of being selected in the mating pool approximately.

Q. 3. Let us assume that the following two parents are participating in uniform crossover of a binary-coded GA.

$$\begin{aligned} Pr1: & 11 \overset{*}{0} 1001 \overset{*}{1} 1100 \overset{*}{0} 01101 \overset{*}{1} 01 \overset{*}{0} \\ Pr2: & 01 \mathbf{1} 1110 \mathbf{0} 1001 \mathbf{0} 01101 \mathbf{0} 110 \end{aligned}$$

Let us also assume that 3<sup>rd</sup>, 8<sup>th</sup>, 13<sup>th</sup>, 19<sup>th</sup> and 22<sup>nd</sup> bit positions counting from the left-most bit have been selected for this crossover. Determine the Children solutions.

Q. 4. In a real-coded GA, let us use polynomial mutation to determine a mutated solution from the original parent solution  $Pr = 21.80$ . Let us assume the random number  $r = 0.3$  for which the perturbation factor  $\bar{\delta} = (2r)^{\frac{1}{q+1}} - 1$ , where the exponent  $q$  is set equal to 4. Assume  $\delta_{\max} = 1.5$ . Determine the mutated solution.

Q.5. Let us consider a constrained optimization problem as given below

$$\begin{aligned} & \text{Minimize } f(x_1, x_2) = x_1^2 + x_2^2 + x_1 + x_2 \\ & \text{subject to } x_1 + x_2 > 4.0 \\ & \quad \quad \quad x_1^2 - x_2^2 + 2x_1x_2 > 16.0 \\ & \text{and } 0.0 \leq x_1, x_2 \leq 10.0 \end{aligned}$$

Solve this problem using the concept of dynamic penalty. Take  $x_1=3.0$ ,  $x_2=5.0$ . Assume the constants  $C=10$ ,  $\alpha=3$ ,  $\beta=4$ . Calculate the value of Penalty term.

Q.6. To minimize the following function:

$$f(x_1, x_2, x_3) = x_1^2 + x_2^2 + 2x_3^2 + x_1 - 2x_2 - x_3$$

in the range of  $-8.0 \leq x_1, x_2, x_3 \leq 8.0$  using Random Walk Method, let us start with an initial solution

$$X_1 = \begin{Bmatrix} x_1 \\ x_2 \\ x_3 \end{Bmatrix} = \begin{Bmatrix} 0.0 \\ 0.0 \\ 0.0 \end{Bmatrix} \text{ and } f_1 = f(X_1). \text{ Assume step length } \lambda = 1.2, \text{ for the first iteration, also take three}$$

random number generated between -1 and 1 as  $\begin{Bmatrix} r_1 \\ r_2 \\ r_3 \end{Bmatrix} = \begin{Bmatrix} 0.62 \\ -0.55 \\ 0.56 \end{Bmatrix}$ . After the first iteration, calculate the value of  $f_2 = f(X_2)$ .

Q. 7. Define Pareto-front of optimal solutions in connection with multi-objectives optimization.